

# Does the model recognise the animal—or the recording?

Acoustic re-identification · 13 datasets · 9 species · 36 neural networks

## 1 Split whole sessions

**+20-51 percentage points**

Random clips raised measured accuracy across 18 count-matched comparisons (ten seeds).  
A higher score does not prove better recognition on a new day.

## 2 Check the background

**71% background · 79% calls**

Balanced accuracy for within-year chiffchaffs.  
Recording context carries substantial information about the assigned identity.

## 3 Separate identity from context

**Timing matters too**

Shared-recording tests support one zebra-finch dataset most clearly. BirdPark lacks same-day same-bird pairs across recordings; time and recording changes cannot be fully separated.

## 4 Let unfamiliar animals arrive

**37% → 46% with another scorer**

Highest median known-call correct acceptance on wild session-disjoint endpoints: mean profile versus nearest clip. Calibration targets 10% strangers; actual test rates can exceed it.

**Test the task you intend to use. Preserve untouched sessions. Calibrate and report stranger errors.**